

FULL-STACK WEB DEVELOPMENT

Session 1:

Project Initialization

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Session Objectives



Architecture

Grasp the core components of Full-Stack Architecture: Frontend, Backend, and Database.



Environment

Set up a professional development environment and project workspace.



Version Control

Initialize a local repository and sync it with GitHub using fundamental Git commands.

01. Theory

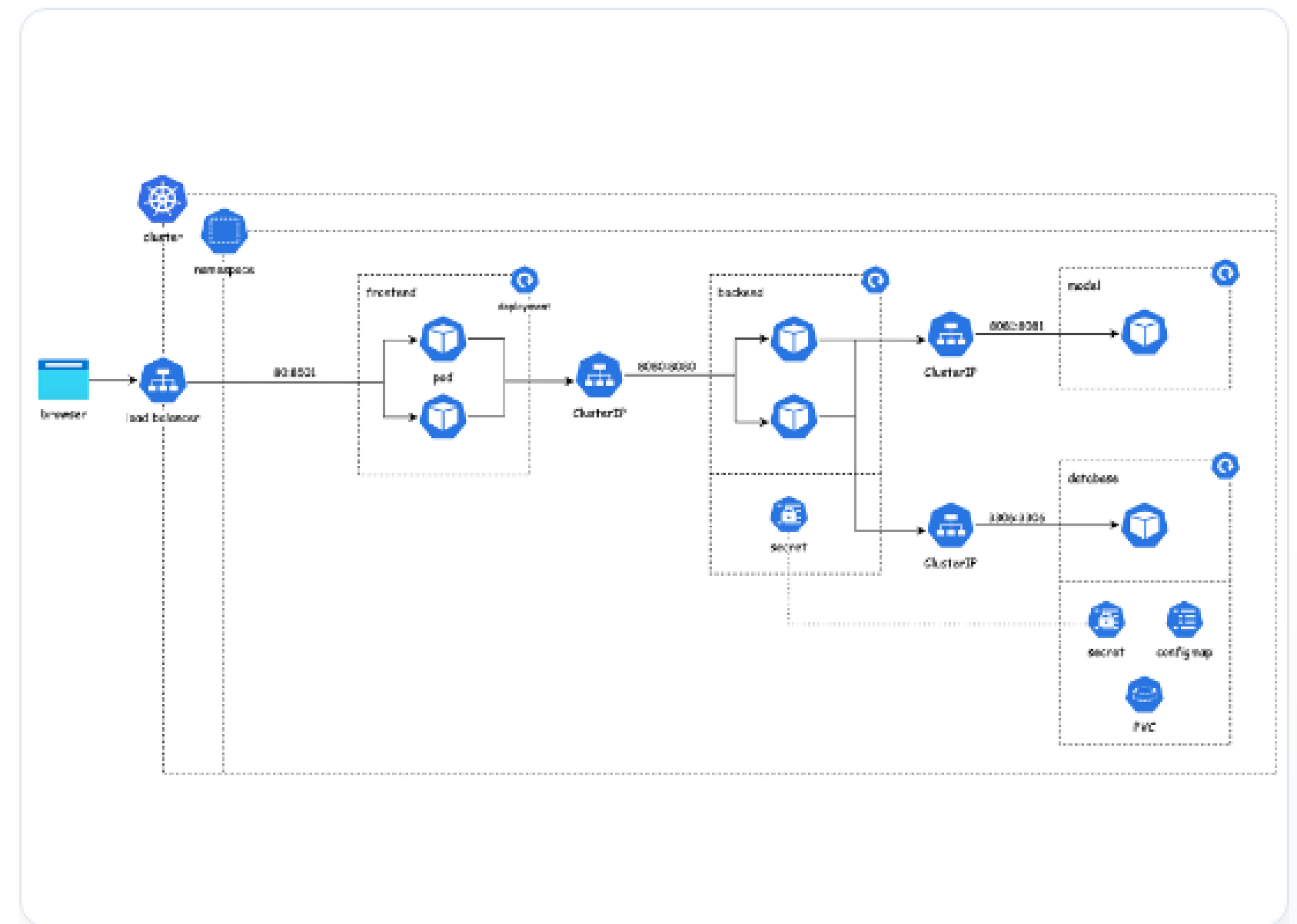
Core Web Architecture & Git Concepts

Full-Stack Web Architecture

How Web Apps Work

Modern web applications are built on a client-server model, where roles are clearly separated for scalability.

- ✓ **Client:** The user interface (Browser).
- ✓ **Server:** The engine processing logic.
- ✓ **HTTP:** The language they use to talk.



The Three Pillars

Frontend

HTML, CSS, and JavaScript (React). This is what users see and interact with. It sends requests to the backend.

Backend

Server-side logic (Node.js). It processes business rules, authentication, and interacts with the database.

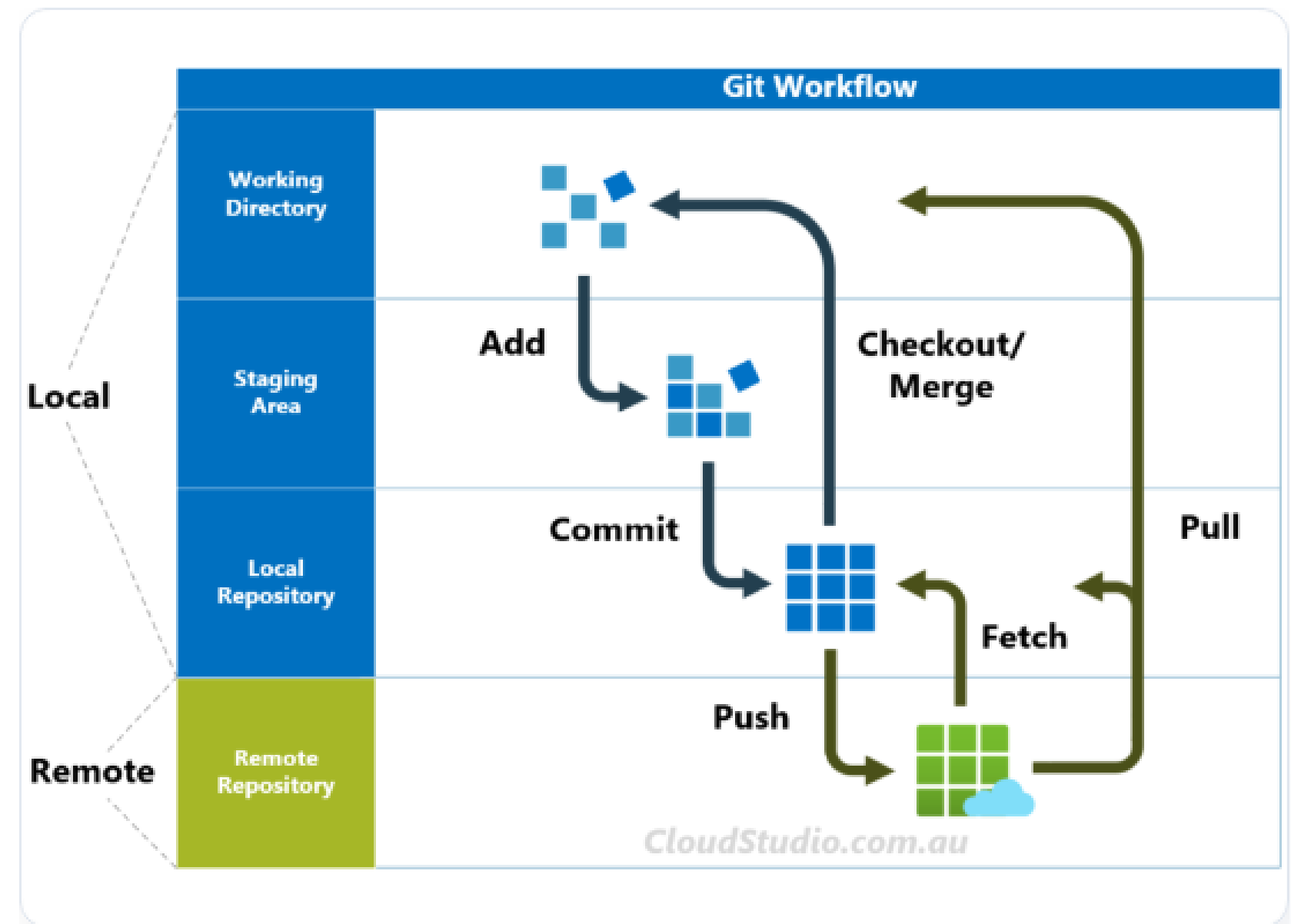
Database

Persistent storage. Ensures your data (users, products, orders) survives when the server restarts.

Git Fundamentals

Version control is essential for tracking changes, collaborating, and reverting code when things go wrong.

-  **Repository:** Your project's central storage.
-  **Commit:** A "snapshot" of your current files.
-  **Push:** Uploading snapshots to GitHub.



02. Hands-on Practice

Setting up the Shophub Workspace

Creating Your Workspace

We will create a root directory named shophub with two distinct sub-projects.



Frontend Folder

Located at shophub/frontend/

Reserved for React applications.



Backend Folder

Located at shophub/backend/

Reserved for Node.js API logic.

Git Command Execution

Step-by-step Initialization

```
$ git init
```

// Creates a local repository

```
$ git add .
```

// Stages all folder changes

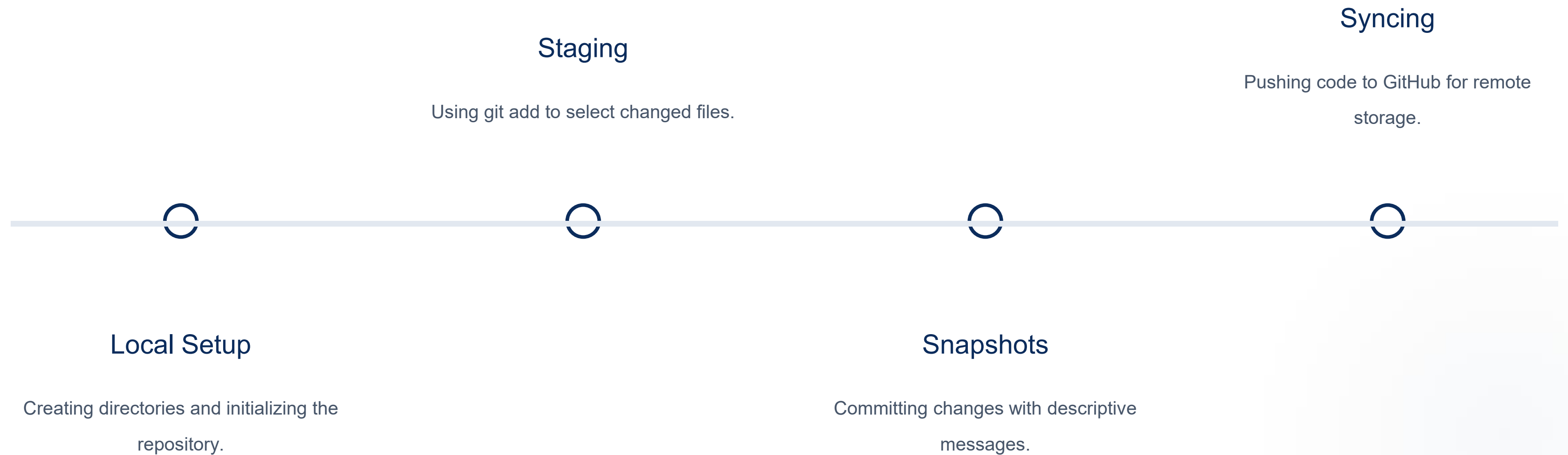
```
$ git commit -m "initial project setup"
```

// Saves the first snapshot

```
$ git push origin main
```

// Syncs local code to the cloud

Development Workflow



Session Outcome

Project Root: shophub/

-  **frontend/** (Web UI)
-  **backend/** (API Services)
-  **.git/** (Version Tracking)

Status: Initialized



Workspace is ready for development.



Thank You